United States Patent	12418794
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Beveridge; Daniel James et al.

Mobile device authentication

Abstract

A desktop is unlocked or locked using a mobile client device, such as a smart phone, tablet, smart watch, etc. The authentication mechanism of the mobile client device, such as fingerprint, facial recognition, voice recognition, username and password, is leveraged for faster, less-cumbersome user authentication on the desktop. In this vein, a client device is added to an authentication agent on the desktop, and the desktop recognizes successful attempts to access the mobile client device as a method of unlocking or locking the desktop.

Inventors: **Beveridge; Daniel James (Apollo Beach, FL), Watts; Blake (St. George, UT), Mu; Jian (BeiJing, CN)**

Applicant: **Omnissa, LLC** (Mountain View, CA)

Family ID: **1000008819457**

Assignee: **Omnissa, LLC (Mountain View, CA)**

Appl. No.: **15/175927**

Filed: **June 07, 2016**

Prior Publication Data

Document Identifier	Publication Date
US 20160342784 A1	Nov. 24, 2016

Related U.S. Application Data

continuation parent-doc US 13217484 20110825 US 9009219 20150414 child-doc US 14686769
continuation-in-part parent-doc US 14686769 20150414 US 10827004 child-doc US 15175927
us-provisional-application US 62211850 20150830
us-provisional-application US 62211736 20150829
us-provisional-application US 61508404 20110715

Publication Classification

Int. Cl.: **H04W12/06** (20210101); **G06F9/451** (20180101); **G06F21/32** (20130101); **G06F21/34** (20130101); **G06F21/35** (20130101); **H04L9/40** (20220101); **H04L67/08** (20220101); **H04L67/75** (20220101)

U.S. Cl.:

CPC **H04W12/06** (20130101); **G06F9/452** (20180201); **G06F21/34** (20130101); **G06F21/35** (20130101); **H04L63/0853** (20130101); **H04L67/08** (20130101); **H04L67/75** (20220501); G06F21/32 (20130101)

Field of Classification Search

CPC: G06F (21/34); G06F (21/35); G06F (21/32); G06F (9/452); H04L (63/0853); H04L (67/36); H04L (67/08); H04W (12/06)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4467421	12/1983	White	N/A	N/A
7117243	12/2005	Peart	N/A	N/A
7181578	12/2006	Guha et al.	N/A	N/A
7184764	12/2006	Raviv et al.	N/A	N/A
7325040	12/2007	Truong	N/A	N/A
7606868	12/2008	Le et al.	N/A	N/A
7669020	12/2009	Shah et al.	N/A	N/A
7941470	12/2010	Le et al.	N/A	N/A
8027354	12/2010	Portolani	N/A	N/A
8112748	12/2011	Pomerantz	N/A	N/A
8234236	12/2011	Beaty	N/A	N/A
8261268	12/2011	Forgette	N/A	N/A
8266311	12/2011	Virdi et al.	N/A	N/A
8271528	12/2011	Wilkinson et al.	N/A	N/A
8356084	12/2012	Yamamoto	N/A	N/A
8396452	12/2012	Matsuoka	455/410	H04W 12/08
8433802	12/2012	Head et al.	N/A	N/A
8494576	12/2012	Bye	455/550.1	G07C 9/00571
8555274	12/2012	Chawla et al.	N/A	N/A
9009219	12/2014	Chawla et al.	N/A	N/A
9075979	12/2014	Queru	N/A	H04W 12/0605
9509676	12/2015	Johnson	N/A	H04W 12/065
2003/0195950	12/2002	Huang et al.	N/A	N/A
2006/0080397	12/2005	Chene et al.	N/A	N/A
2006/0128305	12/2005	Delalat	455/41.2	G08B 13/1418

2007/0174410	12/2006	Croft et al.	N/A	N/A
2007/0180493	12/2006	Croft et al.	N/A	N/A
2007/0186212	12/2006	Mazzaferri et al.	N/A	N/A
2007/0192329	12/2006	Croft et al.	N/A	N/A
2007/0198656	12/2006	Mazzaferri et al.	N/A	N/A
2007/0204011	12/2006	Shaver et al.	N/A	N/A
2007/0260702	12/2006	Richardson et al.	N/A	N/A
2008/0034071	12/2007	Wilkinson et al.	N/A	N/A
2008/0250424	12/2007	Brugiolo et al.	N/A	N/A
2008/0281798	12/2007	Chatterjee et al.	N/A	N/A
2008/0320583	12/2007	Sharma et al.	N/A	N/A
2009/0006537	12/2008	Palekar et al.	N/A	N/A
2009/0094523	12/2008	Treder et al.	N/A	N/A
2009/0216975	12/2008	Halperin et al.	N/A	N/A
2009/0235358	12/2008	Tolba	N/A	N/A
2009/0240904	12/2008	Austruy et al.	N/A	N/A
2009/0276475	12/2008	Ramsey	N/A	H04L 63/083
2010/0070870	12/2009	Halperin et al.	N/A	N/A
2010/0070978	12/2009	Chawla et al.	N/A	N/A
2010/0114867	12/2009	Olston	N/A	N/A
2010/0131654	12/2009	Malakapalli et al.	N/A	N/A
2010/0131949	12/2009	Ferris	N/A	N/A
2010/0191783	12/2009	Mason	N/A	N/A
2010/0199276	12/2009	Umbehocker	N/A	N/A
2010/0211944	12/2009	Kaneda	N/A	N/A
2010/0268813	12/2009	Pahlavan et al.	N/A	N/A
2010/0269152	12/2009	Pahlavan et al.	N/A	N/A
2010/0274784	12/2009	Acharya	N/A	N/A
2011/0004680	12/2010	Ryman	N/A	N/A
2011/0066668	12/2010	Guarraci	N/A	N/A
2011/0099297	12/2010	Hayton	N/A	N/A
2011/0119668	12/2010	Calder et al.	N/A	N/A
2011/0153716	12/2010	Malakapalli et al.	N/A	N/A
2011/0184993	12/2010	Chawla	N/A	N/A
2011/0185292	12/2010	Chawla et al.	N/A	N/A
2011/0185355	12/2010	Chawla et al.	N/A	N/A
2011/0209064	12/2010	Jorgensen et al.	N/A	N/A
2011/0225426	12/2010	Agarwal	713/175	G06F 21/41
2012/0023558	12/2011	Rafiq	726/6	H04L 63/08
2012/0216260	12/2011	Crawford	726/5	G06F 21/31
2013/0018939	12/2012	Chawla et al.	N/A	N/A
2013/0191902	12/2012	Friedl	726/7	G06F 21/41
2014/0189808	12/2013	Mahaffey	726/4	H04L 63/0853
2014/0270127	12/2013	Becker	379/194	H04M 1/67
2014/0308930	12/2013	Tran	455/414.1	H04W 4/001
2014/0372762	12/2013	Flautner	713/173	H04L 63/08
2015/0028996	12/2014	Agrafioti	340/5.82	G06F 21/34
2015/0186636	12/2014	Tharappel	726/8	G06F 21/32
2015/0257004	12/2014	Shanmugam	455/411	H04W 12/065
2015/0295901	12/2014	Woodward	713/168	H04W 12/06

2015/0326578	12/2014	Hsu	726/9	H04L 63/10
2015/0347738	12/2014	Ulrich	726/17	G06F 21/35
2016/0048705	12/2015	Yang	726/34	H04W 12/06
2016/0224779	12/2015	Kitane	N/A	G06F 21/35
2016/0246341	12/2015	Burrell	N/A	G06F 1/26

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
08-212187	12/1995	JP	N/A
2006-081159	12/2005	JP	N/A
2008-511931	12/2007	JP	N/A
2009-230253	12/2008	JP	N/A

OTHER PUBLICATIONS

Clover, Juli, “Apple Working on OS X 10.12 Feature Allowing Macs to Be Unlocked via iPhone's Touch ID”, MacRumors, May 19, 2016, 5 pages. cited by applicant
European Patent Office, International Search Report and Written Opinion for PCT/US2010/046377, transmitted on Jan. 30, 2013, 11 pages. cited by applicant
Quickoffice, Inc., QuickOffice Pro for iPhone and iPod Touch, Version 4.0.0, User Help Guide; Jan. 1, 2010, 42 pages. cited by applicant
Tedd Fox, Citrix Received >> XenApp—Citrix Community, http://community.citrix.com/display/xa/Citrix+Receiver, Aug. 13, 2010. cited by applicant
John McBride et al., Project GoldenGate >> XenApp—Citrix Community, http:/community.citrix.com/display/xa/Project+GoldenGate, Jul. 1, 2011. cited by applicant
Office Action received in copending Australian Patent Application No. 2012284345, Received Feb. 18, 2015, 3 pages. cited by applicant
Japanese Office Action received in Japanese Patent Application No. 2014-520299. Received Dec. 1, 2015. 6 Pages. cited by applicant

---

Primary Examiner: Repsher, III; John T

Attorney, Agent or Firm: Kim & Stewart LLP

Background/Summary

CLAIM OF PRIORITY (1) This application claims the benefit of U.S. Provisional Patent Application Ser. No. 62/211,736, entitled “Mobile Devices in Service of Desktop Uses” filed on Aug. 29, 2015, U.S. Provisional Patent Application Ser. No. 62/211,850, entitled “Mobile Devices in Service of Desktop Uses” filed on Aug. 30, 2015. All of these applications are incorporated herein by reference in their entirety for all intents and purposes. Further, this application is a continuation-in-part to U.S. patent application Ser. No. 14/686,769 entitled “Native Viewer Use for Service Results from a Remote Desktop” filed on Apr. 14, 2015, which is a continuation of U.S. patent application Ser. No. 13/217,484 entitled “Native Viewer Use for Service Results from a Remote Desktop” filed on Aug. 25, 2011, which claims the benefit of U.S. Provisional Patent Application No. 61/508,404 entitled “Native Viewer Use for Service Results from a Remote Desktop” filed on Jul. 15, 2011.

BACKGROUND
(1) Passwords to protect desktop computers are an ongoing barrier to use imposed on users by information technology (IT) departments. Users typically find themselves typing complex

usernames and passwords to unlock their desktops over and over throughout the day, spurred in part by aggressive screen-locking policies meant to enhance security. And most IT policies in today's work environments insist that the passwords to be periodically changed, requiring the users to constantly remember their passwords or, even worse, use easy-to-remember passwords that can easily be hacked.
(2) Although there are biometric authentication devices available for quickly logging into or out of mobile client devices, very few desktops have such capabilities. The cumbersome desktop authentication process is particularly problematic for users who are mobile, or working from more than one terminal during the day. Repetitive unlocking of terminals lead to a frustrating user experience, dissuading users from interacting with the terminals unless the need to do so exceeds the extra steps of the virtual authentication process.
SUMMARY
(3) The disclosed examples are described in detail below with reference to the accompanying drawing figures listed below. The below Summary is provided to illustrate some examples disclosed herein, and is not meant to necessarily limit all examples to any particular configuration or sequence of operations.
(4) Some examples are directed to a method for registering a client device with a remote desktop, to facilitate unlocking the desktop. In that example, a client device, such as a mobile phone, with an authentication mechanism (e.g. facial recognition, voiceprint, fingerprint, etc.) is registered with the remote desktop. The authentication mechanism of the client device then receives an indication from the user. If the indication is accepted by the client device, then the remote desktop is locked or unlocked.
(5) Some examples are directed to a system for unlocking a remote desktop using a client device. In those examples the client device includes an agent which transmits a successful authentication from the client device to the remote desktop. The remote desktop also includes an agent which receives the authentication from the client device, and unlocks the remote desktop in response to the indication.
(6) Some examples are further directed to executable instructions which, when executed by a processor, enable remote authentication of a remote desktop using a client device. The instructions register the client device with an agent operating on the remote desktop. After the client device captures an indication that a user is accessing the client device, the authentication is transmitted to the remote desktop. The remote desktop compares the authentication to valid authentications. If the authentication is valid, the user is able to lock or unlock the remote desktop. But if the authentication is invalid, the remote desktop is locked.
(7) Aspects of the disclosure are not limited to these examples. Variations of these examples, as well as additional examples, are contemplated.
(8) This summary introduces a selection of concepts that are described in more detail below. This summary is not intended to identify essential features. Nor is it meant to limit in any way the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The disclosed examples are described in detail below with reference to the accompanying drawing figures listed below:
(2) FIG. **1** depicts a system for authenticating a user on a remote desktop by an authentication mechanism on a client device.
(3) FIG. **2** depicts a more detailed example of the authentication and registration agents, utilized in mobile device authentication, on the remote desktop.

(4) FIG. **3** illustrates an example user interface for the mobile device authentication techniques disclosed herein.
(5) FIG. **4** depicts a simplified flowchart of a method for mobile device authentication.
(6) FIG. **5** depicts a flowchart of a method for adding a client device and an associated authentication mechanism to a desktop authenticator operating on the remote desktop.
(7) FIG. **6** depicts a more detailed flowchart of a method for mobile device authentication.
(8) FIG. **7** shows a sequence diagram of the operations performed by the client device, the remote desktop, and in some examples the desktop management server during remote device authentication.
(9) FIG. **8** is an example block diagram of a host computing device.
(10) FIG. **9** is a block diagram of an example general virtualized machine architecture environment for executing the described mobile device authentication.
(11) Corresponding reference characters indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

(12) Authentication of a desktop through registered mobile devices is much more convenient for unlocking desktops than lengthy, hard-to-remember, and easily hacked passwords/username combinations. Also, mobile devices are easier to use with one hand than standard desktop keyboards, which helps users whose desktops lock while they talk on the phone or users who need to come and go from their cubicle to attend group meetings. The ability to handle desktop locking/unlocking through the mobile device's simpler and more effective authentication tools reduces the security risks caused by users haphazardly writing and exposing passwords on sticky notes attached to desktop monitors, and allows information technology (IT) personnel to implement aggressive screen locking without burdening users to the same degree.
(13) Some of the examples disclosed herein use a lighter-friction authentication mechanisms of a user's mobile device (e.g., a smart phone, tablet, wearable device, etc.) to lock or unlock the user's desktop screen. In some examples, the user can unlock their desktop screen by authenticating themselves to a mobile device registered with the desktop. For example, a user may step away from the desktop for an extended period of time, causing the desktop to lock, and then, upon return, unlock the desktop screen using a fingerprint scanner of the registered smart phone. Virtually any type of biometric scanner, mobile device passcode, username and password combination, gesture recognition mechanism, or other authentication technique on a mobile device may be used to unlock the desktop screen. Examples of biometric scanners include, without limitation, fingerprint scanners, retina scanners, facial scanners, voice recognition, and the like. Gesture recognition software may integrate with a camera on the mobile device to capture and analyze user movements or with a touch screen on the mobile device to analyze the user's sequential touching of specifically displayed elements.
(14) One particular example uses the authentication performed on a user's smart watch (e.g., Samsung Watch, APPLE WATCH, etc.), either directly or through a paired mobile device (e.g., smart phone or mobile tablet), to unlock or lock the screen of a desktop. To do so, some examples use the smart watch communicatively paired with a mobile device (e.g., smart phone or tablet). In such examples, an indication of the user's authentication on the smart watch is transmitted from the watch to the paired mobile device (e.g., smart phone or tablet) and then from the paired mobile device to the desktop, which in turn uses the smart-watch-captured and paired-watch-relayed authentication indication to unlock or lock the desktop screen. For example, a user may use a fingerprint scanner or watch passcode to authenticate the user on an Apple Watch, and the Apple Watch may transmit that authentication to a paired IPHONE that then transmits the indication of the authentication that occurred on the watch to a desktop for unlocking or locking of the desktop's screen. Such an example uses the Apple Watch and the iPhone working together to control the desktop's locking screen; whereby, the iPhone serves as a relay to send the authentication indication captured on the Apple Watch.

(15) Alternatively, the smart watch itself may act as the mobile device and directly transmit an indication of the user's authentication to the desktop—essentially cutting out the other paired mobile device—and the desktop uses the smart-watch-transmitted authentication indication to unlock or lock the desktop screen. The aforementioned examples enable the user to use smart watches (e.g., the Apple Watch) to remotely lock and unlock their desktop screen. Thus, remote locking and unlocking may be done from a wearable device, either with or without a paired smart phone or other mobile device.
(16) While some examples disclosed herein reference using a fingerprint scanner on the mobile device to unlock the desktop, alternative examples may use other types of biometric unlocking mechanisms. For example, the disclosed unlocking mechanisms may unlock the mobile device and desktop using gesture recognition, facial recognition, eye scanning, voice recognition, signature verification, hand geometry, thermograms and other types of heat patterns, lip motion, gait recognition, ear recognition, skin reflection, keystroke recognition, or other type of biometric recognition technique (collectively “biometric unlocking mechanisms”).
(17) Additionally, the disclosed embodiments may be used across multiple operating systems (OSes) and are not limited to just one particular OS ecosystem. For example, some of the disclosed embodiments may use the biometric sensor of an APPLE® brand iPHONE® to unlock a WINDOWS® brand OS on a desktop, or a fingerprint scanner of a SAMSUNG® brand GALAXY® phone may be used to unlock an APPLE® brand OS X on a remote desktop. In other words, the disclosed embodiments enable desktop unlocking when the mobile device is executing a different OS than the desktop.
(18) The disclosed mobile desktop authentication system increases the accessibility of desktop systems. Users are not forced to type in cumbersome passwords, but are able to rely on biometric markers or more-commonly-entered mobile device passcodes or usernames and password combinations to access desktops. This provides an additional layer of security.
(19) Having described various examples of this disclosure, attention is directed to the architecture and individual devices configured to perform the desktop authentication through mobile authentication capabilities previously discussed.
(20) FIG. **1** depicts a system for authenticating a user on a remote desktop by an authentication mechanism on a client device. Looking at the accompanying figures, FIG. **1** depicts an exemplary environment for providing desktop services. Users may use a mobile device, which is shown and referred to as client device **104**, to interact with a desktop management server **102** to access a user's remote desktop **106**. Client device **104** may use a client services interface to have services, such as authentication, performed by remote desktop **106**. In some example, data related to authentication collected by the client device **104** is authorized authentication data or an authorized indication. In some examples, this may include any of the aforesaid biometric unlocking mechanisms, e.g., gesture recognition, facial recognition, eye scanning, voice recognition, signature verification, hand geometry, thermograms and other types of heat patterns, lip motion, gait recognition, ear recognition, skin reflection, keystroke recognition, or other type of biometric recognition technique. The data used to authenticate a user is referred to as authentication data or an “indication.” An indication is either authorized (e.g., the indication is known and indicates authorized use) or unauthorized. The client services interface may be part of a remote desktop solution allowing access to remote desktop **106**.
(21) Client device **104** may send authentication requests to remote desktop **106**. Remote desktop **106** may be a physical computer system or a virtualized computer system. Physical desktops may be actual physical machines being controlled remotely. Virtualized desktops may export user interfaces, e.g., keyboard and mouse input, audio and visual output, to the user from virtual machines (VMs) running remotely (in the datacenter) or locally on client device **104**, or, in some cases, using a hybrid.
(22) Desktop management server **102** may be an interface layer that includes desktop management

software with various functions. Desktop management server **102** (see FIG. **9**) includes a connection broker **923**, which allows a remote user to select a type of remote desktop **106** (e.g., remote desktop client or client services interface) and initiate a desktop session, or a connection to remote desktop **106** using client device **104**. Client device **104** may be a mobile device, such as a cellular phone, smart phone, tablet, personal computer (PC), laptop or other computing device that may have a different form factor than a remote desktop **106**. For example, remote desktop **106** may virtualize a screen size that is different from the size of the display provided by access device **104**. Additionally, client device **104** may be running a different operating system (OS) with different application characteristics and application programming interfaces (APIs) than remote desktop **106**.
(23) Client device **104** includes a desktop agent **108** that communicates with desktop management server **102** or directly to remote desktop **106**, which may be hosted remotely or locally by a Type **2** hypervisor running on client device **104**. Desktop agent **108** may send service requests to have services performed by the user's remote desktop **106**. In an example, a side channel for a remote desktop session may be used to send the requests and receive the result sets. This side channel may be used instead of a channel used to send authentications of remote desktop **106** using client device **104**. Desktop management server **102** allows non-display traffic to be sent from desktop agent **108** to remote desktop **106** in the side channel. This process is described in more detail below.
(24) Proxy services agent **206** outputs a result set. The result set may include the result of an authentication attempt on remote desktop **106**. Any suitable protocol or API can be implemented for message transmission between client device **104** and desktop management server **102**. For example, various web services protocols can be used, and the message payloads may be encrypted and/or compressed.
(25) The user may be authenticated before accessing remote desktop **106**. A remote desktop gateway **118** is used to authenticate the user. In one example, a new client services interface (not illustrated) is launched by a user to have services performed. The client services interface may be similar to a remote desktop client in terms of establishing connectivity to remote desktop **106**. Although the authentication is described with respect to authenticating the remote desktop **106**, in some embodiments, an independent authentication process may be used for establishing a session for having services performed by remote desktop **106** through the remote desktop gateway **118**. When the client services interface on client device **104** is launched, client device **104** provides credentials to remote desktop gateway **118**. For example, enterprise credentials are provided to authenticate the access for the user to remote desktop **106**. A gateway authenticator **304** authenticates the credentials.
(26) Gateway authenticator **304** may use different methods of authenticating the credentials. In one example, based on the administrative policy, the user may be asked to use a two-factor authentication. For example, the user may be required to first provide a Rivest, Shamir, and Adelman (RSA) SECURID® or another token and then provides active directory credentials. In some examples, the administrative policy controls the access to the remote desktop **106** based on criteria established by an administrator. As an example, the client device **104** is prevented from authenticating the remote desktop **106** without a new token, issued daily to authorized client devices **104** by the remote desktop gateway **118**. In another example, the remote desktop gateway **118** prevents client device **104** from authenticating remote desktop **112** during certain hours (e.g., after hours of business operation). This authentication process is known and need not be described further.
(27) In alternative embodiments, unlocking of the remote desktop **106** is performed by the desktop agent **108** acting as a pass-through agent for providing identification of the remote desktop **106** by way of a reverse-proxy agent or by way of integration with the desktop agent **108**, and thereafter be used to unlock a virtual desktop session. In one specific example, the desktop agent **108** may include a VMWARE HORIZON® branded client that passes through an identifier (e.g., username, media access control (MAC), universally unique identifier (UUID), or the like) of the remote

desktop **106** along with the unlocking indication to a cloud service that uses both to unlock a virtual desktop session in a virtual desktop infrastructure (VDI). Going further, in one example, a scenario may involve a user on a home personal computer with a VMWARE HORIZON® branded client accessing a VDI session in a corporate data center. The VMWARE HORIZON® branded client (i.e., the desktop agent **108**) may unlock the user's local personal computer and a VDI home screen displayed within a HORIZON VIEW® branded client to access a VDI session in the corporate data center. Thus, the desktop client **108** may be used to not only unlock the user's local personal computer, but also to unlock a VDI home screen. This convenient authentication capability may also be made available within the desktop agent **108** and paired with other virtual desktop capabilities, thereby offering the user a comprehensive augmentation of the desktop experience.
(28) FIG. **2** depicts a more detailed example of the authentication and registration agents, utilized in mobile device authentication, on the remote desktop **106**. Within the desktop authenticator **112** are various plug-ins, such as the registration agent **114** a registration agent **114** and an authentication agent **116**, within a proxy services agent **206**. The proxy services agent **206** also includes a command queue **204** that stores various instructions for remote desktop authentication on the client device **104**. As an example, the command queue **204** includes instructions to lock the client device **104** from attempting mobile device authentication upon receiving an invalid unlock attempt. In an example where an invalid unlock attempt is received, an administrator may further be alerted as to the invalid attempt. Additional or alternative plug-ins may be used in various examples. Examples discussed below reference the remote desktop **106** interacting with the client device **104** to enable authentication of the remote desktop **106**.
(29) In one example, registration agent **114** registers the client device **104** for locking and unlocking the screen of the remote desktop **106**. Authentication agent **116** issues commands to lock and unlock the remote desktop **106** upon notice of successful user authentication (e.g., fingerprint scanning, mobile device passcode entry, voice recognition, retina scan, username and password, gesture recognition, etc.) on the client device **104**.
(30) Proxy services agent **206** receives service requests for locking/unlocking the remote desktop **106** through remote authentication by the user. Proxy services agent **206** performs the requested services, optionally, by performing system calls to the operating system running on remote desktop **106**. For example, the remote desktop **106** screen may be locked or unlocked by accessing system functions of the operating system on remote desktop **106**. Also, applications running on remote desktop **106** may also be queried to perform the services. A result set gathered by proxy service agent **110** and then returned to client device **104**. The type of result set may depend on the service performed.
(31) Registration agent **114** registers the client device **104** (e.g., mobile devices) with the remote desktop **106** for mobile device authentication. This process is described in more detail in FIG. **5**. To do so, registration agent **114** may identify and store a particular address (e.g., MAC, Internet Protocol (IP), or other device identifiers) of the client device **104** to be used for authentication of the remote desktop **106**. Additionally or alternatively, registration agent **114** may register the client device **104** for unlocking and locking the remote desktop **106**.
(32) Authentication agent **116** issues lock and unlock commands for the remote desktop **106** upon receipt of authentication command from the client device **104**, which is registered with the registration agent **114**, to provide remote unlocking functionality. For example, the desktop agent **108** on the client device **104** may be configured to transmit an authentication command upon authentication of a user on the client device **104** (e.g., successful scanning of fingerprint, username and password, facial recognition, input of mobile device passcode, etc.) that causes the authentication agent **116** to unlock or lock the screen of the remote desktop **106**. In an example, the desktop agent **108** on the client device **104** issues the authentication command to lock/unlock the remote desktop **106** using a Representational State Transfer (REST) API. Other examples may issue the authentication command in alternative ways.

(33) FIG. **3** illustrates an example user interface for the mobile device authentication techniques disclosed herein. FIG. **3** shows a user being authenticated to a client device **104** using a fingerprint scanner, and the authentication on the client device **104** is used to remotely control the locking or unlocking of the remote desktop **106**. The client device, in some examples, is a cellular telephone, smart phone, smart watch, tablet, fob, or any other device able to authenticate a user. In some examples the client device includes technology to recognize: fingerprint, username and password, facial recognition, input of mobile device passcode, voiceprint, pattern or sequence, etc. In other examples the client device is wirelessly enabled, and carried by a user. The client device itself, in that example, acts as the authentication. The remote desktop **106** includes traditional desktop computing systems, as well as virtual desktops. The remote desktop, in some examples, is connected to a network, or it is standalone. In some examples the remote desktop **106** is a terminal, upon which a virtual desktop or virtual machine is instantiated upon request or upon unlocking. Alternatively, a virtual machine is suspended or resumed based upon a mobile device authentication attempt.
(34) FIG. **4** depicts a simplified flowchart of a method for mobile device authentication. A client device **104** is initially registered with the remote desktop **106** for using the client device's authentication mechanism (e.g., biometric scan, passcode, gesture, username and password, etc.) to unlock the remote desktop, as indicated at block **402**. The registration is described in more detail in FIG. **5**. An indication that a user performed an unlocking mechanism on the client device **104** is received, as shown at block **404**. The client device **104** unlocking mechanism is then used to either lock or unlock the remote desktop **106**, as indicated at block **406**, and described in more detail in FIG. **6**. In other words, the remote desktop **106** is unlocked or locked based on the user being authenticated to the client device **104**. In some examples the indication is an authorized indication (recognized as valid either by the client device **104** or the remote desktop **106**). Alternatively, an indication is unauthorized, and not recognized by either device.
(35) FIG. **5** depicts a flowchart of a method for adding a client device and an associated authentication mechanism to a desktop authenticator **112** operating on the remoted desktop **106**. At **502** the registration agent **114** is launched, executed, or started. In some examples, the registration agent **114** is running, already. At **504**, the registration agent **114** optionally searches for locally or remotely connected client devices **104**. In some examples, the registration agent **114** locates any client devices **104** which is wirelessly available, or which the registration agent **114** can “see” on a wireless network. Alternatively, the registration agent **114** searches the remote desktop **106** for details of known client devices **104**. As an example, a client device **104** associated with an iTunes account is located by the registration agent **114**. If client devices **104** are detected at **506**, then the registration agent **114** suggests adding the detected client devices **104** at **508**. In some examples, a list is presented to a user permitting the user to select client devices **104** to add to the authentication agent **116**. At **510**, if one or more of the suggested client devices **104** are added, then those client devices **104** are added to the authentication agent **116** at **512**.
(36) Adding the client device **104** to the authentication agent **116** includes, in some example, adding information about the authentication mechanism of the client device **104**, and the authorized authentication data or authorized indication associated with the client device **104**. In some examples, this may include facial recognition, voice recognition, fingerprint, username and password, gesture, or other identifying information. The data used to authenticate a user is referred to as authentication data or an “indication.” An indication is either authorized (e.g., the indication is known and indicates authorized use) or unauthorized. Alternatively, in some examples, the authentication agent **116** stores no specific information about the authentication data associated with the client device **104** and only stores whether or not the client device **104** received the valid, correct authentication data. The registration agent **114** is also terminated upon registering the client device **104** with the authentication agent **116** at **512**.
(37) If, at **506** no client devices **104** are detected, or if none of the suggested client devices are

selected at **510**, then a user may be prompted to manually add or register a client device **104** at **514**. Manually adding a client device **104** includes, in some examples, adding information associated with the client device **104**, such as a telephone number, a mobile identification number (MIN), mobile subscription identification number (MSIN), mobile device identifier, device identifier, unique device identifier (UDID), etc. At **516**, if the client device **104** is manually added, then the client device **104** is registered with the authentication agent **116** at **512**, and the registration agent **114** terminates. If no client device **104** is added at **516**, then the registration agent **114** terminates at **518**.

(38) FIG. **6** depicts a more detailed flowchart of a method for mobile device authentication. At **602** the authentication mechanism of the client device **104** receives an indication or an unlocking indication. Examples of the authentication mechanism include, without limitation, a valid fingerprint scan, a retina scan, username and password, a passcode, voice recognition, a facial scan, a recognized gesture, or any other kind of authentication mechanism. Alternatively, a sequence of events acts as an authorized indication, if it satisfies a credential policy. As an example, if the client device detects a location change from “home” to “work,” and then the client device detects that it is entering “the office,” this sequence of events may act as an authorized indication to trigger the mobile desktop authentication system, either solely or in combination with the disclosed authentication mechanisms. Performing this sequence of events may identify the user, in accordance with the credential policy. Similarly, authentication may be disallowed when the user is not within a particular proximity to the remote desktop (e.g., within 5 meters). Thus, the location of the user may be used either to trigger a user to perform the authentication mechanism on the client device or as a condition for accepting a user's authentication.

(39) If the indication is authorized on the client device **104**, then in some examples the credentials of the client device **104** are transmitted to the gateway authenticator **120**. The transmitted credentials include, as an example, a token previously provided by the desktop management server **102** to indicate that the client device **104** is authorized to access the enterprise system. If the authorized indication is not detected, then access is denied at **614**. In some examples an administrator is notified if access is denied.

(40) In some examples an unauthorized indication results in locking an unlocked remote desktop **106**, locking the client device **104**, or disabling the mobile device authentication system. In an example where credentials are transmitted to the gateway authenticator **120**, the gateway authenticator **120** verifies that the client device **104** is authorized to access the enterprise system at **606**. If the client device **104** is not authorized, then access is denied at **614**. In some examples, the gateway authenticator **120** compares the client device **104** and the access details with an administrative policy such as an enterprise credential policy to determine whether they match known or accepted access details. The administrative policy includes, for example, rules about access levels, access frequency, user location, etc. If the client device **104** is authorized for enterprise access, or if operation **606** is not required, the receipt of the authorized indication is transmitted to the desktop authenticator **112** at **608**. In some examples, the client device **104** is authorized for enterprise access based on the time of day (e.g. during their shift), the location of the client device (access not authorized if the client device is not within a range, such as 5 meters, of the remote desktop), etc. If the client device **104** satisfies all administrative policies locally on the remote desktop **106** at **612** then the remote desktop **106** is locked or unlocked as requested at **612**. Otherwise access to the desktop through mobile device authentication is denied at **616**.

(41) FIG. **7** shows a sequence diagram of the operations performed by the client device **104**, the remote desktop **106**, and in some examples the desktop management server **102** during remote device authentication. Initially the registration agent **114** is launched by the remote desktop **106**. The operations involved in launching the registration agent **114**, and adding a client device **104** to the authentication agent **116**, are described in more detail in FIG. **4**. Optionally, the registration agent suggests adding client devices **104** to the authentication agent **116**. At least one client device

104 is added to the authentication agent **116**. In some examples, the client device **104** is also added to the gateway authenticator **120**. In examples utilizing the gateway authenticator **120**, the client device **104** is provided with a token or other credential to use to access the desktop management server **102**.

(42) The authentication mechanism on the client device receives an indication (e.g., an attempt is made to access the client device **104** through the authentication mechanism). If the indication is authorized, then the authorized indication is passed to the remote desktop **106**, and in some examples the desktop management server **102**. Based, in some examples, upon administrative policies, the desktop management server **102** allows or denies access to the remote desktop by the mobile device authentication system. The remote desktop **106** is then locked or unlocked based upon the authorized indication.

(43) FIG. **8** is a block diagram of an example host computing device **800**. Host computing device **800** includes a processor **802** for executing instructions. In some examples, executable instructions are stored in a memory **804**. Memory **804** is any device allowing information, such as executable instructions and/or other data, to be stored and retrieved. For example, memory **804** may include one or more random access memory (RAM) modules, flash memory modules, hard disks, solid state disks, and/or optical disks.

(44) Host computing device **800** may include a user interface device **810** for receiving data from a user **808** and/or for presenting data to user **808**. User **808** may interact indirectly with host computing device **800** via another computing device such as a device running VMware's vCenter Server or other management device. User interface device **810** may include, for example, a keyboard, a pointing device, a mouse, a stylus, a touch sensitive panel (e.g., a touch pad or a touch screen), a gyroscope, an accelerometer, a position detector, and/or an audio input device. In some examples, user interface device **810** operates to receive data from user **808**, while another device (e.g., a presentation device) operates to present data to user **808**. In other examples, user interface device **810** has a single component, such as a touch screen, that functions to both output data to user **808** and receive data from user **808**. In such examples, user interface device **810** operates as a presentation device for presenting information to user **808**. In such examples, user interface device **810** represents any component capable of conveying information to user **808**. For example, user interface device **810** may include, without limitation, a display device (e.g., a liquid crystal display (LCD), organic light emitting diode (OLED) display, or “electronic ink” display) and/or an audio output device (e.g., a speaker or headphones). In some examples, user interface device **810** includes an output adapter, such as a video adapter and/or an audio adapter. An output adapter is operatively coupled to processor **802** and configured to be operatively coupled to an output device, such as a display device or an audio output device.

(45) Host computing device **800** also includes a network communication interface **812**, which enables host computing device **800** to communicate with a remote device (e.g., another computing device) via a communication medium, such as a wired or wireless packet network. For example, host computing device **800** may transmit and/or receive data via network communication interface **812**. User interface device **810** and/or network communication interface **812** may be referred to collectively as an input interface and may be configured to receive information from user **808**.

(46) Host computing device **800** further includes a storage interface **816** that enables host computing device **800** to communicate with one or more data storage devices, which store virtual disk images, software applications, and/or any other data suitable for use with the methods described herein. In example examples, storage interface **816** couples host computing device **800** to a storage area network (SAN) (e.g., a Fibre Channel network) and/or to a network attached storage (NAS) system (e.g., via a packet network). The storage interface **816** may be integrated with network communication interface **812**.

(47) FIG. **9** is a block diagram of an example general virtualized machine architecture environment for executing the described mobile device authentication. The following architecture may be used

in a remote desktop session and may be enhanced to use the client services interface to have services performed and return result sets for display using native viewers. As shown in FIG. **9**, in traditional use, remote users, for example users **905***a* and **905***b*, may access centrally-managed remote desktops **106**, such as those implemented by virtual machines **935** running in a datacenter, using network **910** (e.g., a local area network, or other private or publically accessible wide area network, such as the Internet) through any number of different types of devices. These virtual machines (VMs) **935** are complete computation environments, containing virtual equivalents of the hardware and system software components of a physical system as described above, and are typically implemented by an extensive virtualization infrastructure, which includes a variety of software and hardware components.

(48) Remote access to remote desktops is generally provided to client devices through a desktop management server **102**. Desktop management server **102** provides access to remote desktops by the remote user devices, and manages the corresponding virtual machines through communications with a software interface **932** of a Virtual Machine Management Server (VMMS) **930**. The Virtual Machine Management Server (VMMS) **930** is responsible for provisioning and maintaining the multitude of Virtual Machines (VMs) **935** implemented across potentially a multitude of physical computers, such as host computing devices **800**. When a user wishes to access an existing virtual machine, the user establishes a connection through the desktop management server **102**, and a remote desktop is presented (as a user interface) on the user's client device, through which communications are made with the underlying virtual machine. Additionally, the virtual machine may include a proxy services agent **206** as described above with reference to FIGS. **1** and **4** to perform service requests.

(49) In the example shown, each physical computer, for example computer **800** contains the underlying hardware **940**, virtualization software (here a hypervisor **980**), and one or more virtual machines, for example VM **935**.sub.1 and VM **935**.sub.2, which each contain Agent Software (guest system software) labeled here as “A” in each VM box. The Agent Software is typically responsible for connecting each VM to the desktop management server **102** and manages each desktop connection. It typically notifies the desktop management server **102** upon each login, logoff, and disconnect. The Agent Software also provides support for remote devices such as universal serial bus (USB) devices, etc. The Agent Software may also be enhanced to include proxy services agent **206** and service plug-ins **112**.

(50) The VMMS **930** typically manages pools of compute resources used to run virtual machines on a set of clusters typically containing multiple servers with CPUs, memory, and communications hardware (network). A virtual computer (a virtual machine or VM), when active, consumes physical computer resources and is managed by a hypervisor layer, such as hypervisor **980** running on physical computer **800**. The hypervisor manages physical resources as well as maintains virtual-to-physical hardware mappings. The Software Interface **932** running on the VMMS **930** communicates with these hypervisors (e.g., hypervisor **980**) to provision and manage each VM **935**. For example, according to traditional virtualization techniques, when a remote user (e.g., user **905***a*) requests access to a particular existing desktop, the desktop management server **102** (through its software **925**), communicates with the VMMS through its software interface **932** to start the corresponding VM **935** executing on an appropriate physical computer, and to relay the user interface exported by the VM **935** to the remote user so that the user can interact with the remote desktop **106**. In some instances (e.g., according to administrator policies), when the desktop is exited, or otherwise shutdown, the desktop management server **102** communicates with the VMMS **930** to save the VM image to the datastore **970** as appropriate and to de-allocate physical and VM system resources as needed.

(51) In general, the VMMS Server **930** provides interfaces **932** to enable other programs, such as the Pool Manager **924**, to control the lifecycle of the various virtual machines that run on a hypervisor. In one example of an existing virtualization infrastructure provided by VMware Inc.,

desktop management server **102** includes an Administrative Console **921**, an Inventory Manager **922**, a Connection broker **923**, and a Pool Manager **924**. The Connection broker **923** allows a remote user, such as remote user **905***a*, through client device **104**, to initiate a desktop session with an assigned VM **935** or to access an existing connection to VM **935**. Connection broker **923** may also be enhanced to include remote desktop gateway **118**.

(52) The Inventory Manager **922** maintains a mapping of different user belongings in the system. For example, user may be entitled to certain applications; may have access to more than one desktop, etc. The Inventory Manager **922** also keeps track of the running remote desktops in the system. The mappings may be stored using any number of mechanisms, including using one or more directory servers **915** accessible through network **910**.

(53) The Pool Manager **924** component manages the complete lifecycle of remote desktops. Desktops in a pool are grouped together based on similar software requirements. Desktop Administrators create logical desktops groups (desktop pools) that are provisioned typically from the same base image, including the Agent Software. For example, a desktop pool may include virtual machines that run the same set of software applications and run the same operating system. As yet another example, a desktop pool may contain a set of cloned virtual machines that are identical in every aspect but are customized to include unique identity that includes for example, a unique computer name, IP/MAC Address, Domain membership, Software license serial numbers, OS specific security identifiers among other things. The base image can be a virtual machine or a template virtual machine that is created and/or managed by the VMMS **930**.

(54) The software state of all the virtual machines **935** in a desktop pool may be persistent or non-persistent. Persistent desktops maintain the state of the files or applications stored inside the virtual machines. Non-Persistent desktops are stateless desktops; the desktop state is restored to the original state after every user session. In some cases, the Desktop Administrator can define how frequently the “revert to golden state” operation should be performed. The restore to pristine image or revert to golden state operation can also be scheduled to occur based on certain conditions.

(55) The Administrative Console **921** typically provides a user interface for a Desktop Administrator to manage the configuration of desktop pools, define user access policies, manage ongoing maintenance, software installed in the desktops, etc.

(56) The Directory Server **915** stores the persistent state required for managing the remote desktops. For example, the VMs in a desktop pool maybe associated with one or more users. The user identifiers for a pool may be stored in the directory server **915**. The users may also be referenced through an external directory server such as—Microsoft Active Directory, Novell eDirectory, IBM Tivoli Directory Server, etc.

(57) The illustrated virtualization architecture is shown using a hypervisor-centric model. Alternative examples may control some or all of the VMs **935** in a virtualization architecture that uses containers instead of hypervisors. Yet, other examples may manage the depicted VM environment with unikernels. Thus, the illustrated virtualization architecture is but only one configuration for implementing the various examples disclosed herein.

(58) Examples are not limited to being executed in virtualized environments, however. The disclosed examples may be run simply between desktop, server, and client mobile devices in a public or private network environment.

(59) In some examples, executable instructions are stored in a memory. Memory is any device allowing information, such as executable instructions and/or other data, to be stored and retrieved. For example, memory may include one or more random access memory (RAM) modules, flash memory modules, hard disks, solid state disks, and/or optical disks.

Additional Examples

(60) In some examples, a system includes a desktop agent **108** that cooperates with a mobile agent to provide a range of augmented services to desktop users, allowing the mobile device to function as an extension of the desktop for designated tasks. In some examples, the multiple helper devices

are registered simultaneously to the desktop for purposes of serving as augmentations to user productivity or enhancements to user experience. In some examples, a user can authenticate to a remote desktop **106** session using the touch-ID capability of a mobile device such as an Apple iPad or iPhone, or wearable watch. In some examples, a user may lock or unlock the desktop's screen using Touch-ID.

(61) In some examples, a non-desktop device such as a touch-ID fingerprint device is used to authenticate and unlock the desktop operating system.

(62) The various examples described herein may employ various computer-implemented operations involving data stored in computer systems. For example, these operations may require physical manipulation of physical quantities—usually, though not necessarily, these quantities may take the form of electrical or magnetic signals, where they or representations of them are capable of being stored, transferred, combined, compared, or otherwise manipulated. Further, such manipulations are often referred to in terms, such as producing, identifying, determining, or comparing. Any operations described herein that form part of one or more examples may be useful machine operations. In addition, one or more examples also relate to a device or an apparatus for performing these operations. The apparatus may be specially constructed for specific required purposes, or it may be a general purpose computer selectively activated or configured by a computer program stored in the computer. In particular, various general purpose machines may be used with computer programs written in accordance with the teachings herein, or it may be more convenient to construct a more specialized apparatus to perform the operations. The various examples described herein may also be practiced with other computer system configurations including hand-held devices, microprocessor systems, microprocessor-based or programmable consumer electronics, minicomputers, mainframe computers, and the like.

(63) The examples illustrated and described herein as well as examples not specifically described herein but within the scope of aspects of the disclosure constitute exemplary means for mobile desktop authorization. For example, the elements illustrated in the figures, such as when encoded to perform the operations illustrated in the figures, constitute exemplary means for implementing the operations recited in the Claims and/or otherwise described herein.

(64) One or more examples may be implemented as one or more computer programs or as one or more computer program modules embodied in one or more computer readable storage media. The term computer readable storage medium refers to any data storage device that can store data which can thereafter be input to a computer system—computer readable media may be based on any existing or subsequently developed technology for embodying computer programs in a manner that enables them to be read by a computer. Examples of a non-transitory computer readable medium include a hard drive, network attached storage (NAS), read-only memory, random-access memory (e.g., a flash memory device), a CD (Compact Discs)—CD-ROM, a CDR, or a CD-RW, a DVD (Digital Versatile Disc), a magnetic tape, and other optical and non-optical data storage devices. The computer readable medium can also be distributed over a network coupled computer system so that the computer readable code is stored and executed in a distributed fashion.

(65) In addition, while described virtualization methods have generally assumed that virtual machines present interfaces consistent with a particular hardware system, persons of ordinary skill in the art will recognize that the methods described may be used in conjunction with virtualizations that do not correspond directly to any particular hardware system. Virtualization systems in accordance with the various examples, implemented as hosted examples, non-hosted examples or as examples that tend to blur distinctions between the two, are all envisioned. Furthermore, various virtualization operations may be wholly or partially implemented in hardware.

(66) Many variations, modifications, additions, and improvements are possible, regardless the degree of virtualization. The virtualization software can therefore include components of a host, console, or guest operating system that performs virtualization functions. Plural instances may be provided for components, operations or structures described herein as a single instance. Finally,

boundaries between various components, operations and data stores are somewhat arbitrary, and particular operations are illustrated in the context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within the scope of the invention(s). In general, structures and functionality presented as separate components in exemplary configurations may be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component may be implemented as separate components.

(67) The operations described herein may be performed by a computer or computing device. The computing devices communicate with each other through an exchange of messages and/or stored data. Communication may occur using any protocol or mechanism over any wired or wireless connection. A computing device may transmit a message as a broadcast message (e.g., to an entire network and/or data bus), a multicast message (e.g., addressed to a plurality of other computing devices), and/or as a plurality of unicast messages, each of which is addressed to an individual computing device. Further, in some examples, messages are transmitted using a network protocol that does not guarantee delivery, such as User Datagram Protocol (UDP). Accordingly, when transmitting a message, a computing device may transmit multiple copies of the message, enabling the computing device to reduce the risk of non-delivery.

(68) By way of example and not limitation, computer readable media comprise computer storage media and communication media. Computer storage media include volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer readable instructions, data structures, program modules or the like. Computer storage media are tangible, non-transitory, and are mutually exclusive to communication media. In some examples, computer storage media are implemented in hardware. Exemplary computer storage media include hard disks, flash memory drives, digital versatile discs (DVDs), compact discs (CDs), floppy disks, tape cassettes, and other solid-state memory. In contrast, communication media typically embody computer readable instructions, data structures, program modules, or the like in a modulated data signal such as a carrier wave or other transport mechanism, and include any information delivery media.

(69) Although described in connection with an exemplary computing system environment, examples of the disclosure are operative with numerous other general purpose or special purpose computing system environments or configurations. Examples of well-known computing systems, environments, and/or configurations that may be suitable for use with aspects of the disclosure include, but are not limited to, mobile computing devices, personal computers, server computers, hand-held or laptop devices, multiprocessor systems, gaming consoles, microprocessor-based systems, set top boxes, programmable consumer electronics, mobile telephones, network PCs, minicomputers, mainframe computers, distributed computing environments that include any of the above systems or devices, and the like.

(70) Examples of the disclosure may be described in the general context of computer-executable instructions, such as program modules, executed by one or more computers or other devices. The computer-executable instructions may be organized into one or more computer-executable components or modules. Generally, program modules include, but are not limited to, routines, programs, objects, components, and data structures that perform particular tasks or implement particular abstract data types. Aspects of the disclosure may be implemented with any number and organization of such components or modules. For example, aspects of the disclosure are not limited to the specific computer-executable instructions or the specific components or modules illustrated in the figures and described herein. Other examples of the disclosure may include different computer-executable instructions or components having more or less functionality than illustrated and described herein.

(71) Aspects of the disclosure transform a general-purpose computer into a special-purpose computing device when programmed to execute the instructions described herein.

(72) At least a portion of the functionality of the various elements illustrated in the figures may be

performed by other elements in the figures, or an entity (e.g., processor, web service, server, application program, computing device, etc.) not shown in the figures.
(73) In some examples, the operations illustrated in the figures may be implemented as software instructions encoded on a computer readable medium, in hardware programmed or designed to perform the operations, or both. For example, aspects of the disclosure may be implemented as a system on a chip or other circuitry including a plurality of interconnected, electrically conductive elements.
(74) The order of execution or performance of the operations in examples of the disclosure illustrated and described herein is not essential, unless otherwise specified. That is, the operations may be performed in any order, unless otherwise specified, and examples of the disclosure may include additional or fewer operations than those disclosed herein. For example, it is contemplated that executing or performing a particular operation before, contemporaneously with, or after another operation is within the scope of aspects of the disclosure.
(75) These and other variations, modifications, additions, and improvements may fall within the scope of the appended claims(s). As used in the description herein and throughout the claims that follow, “a,” “an,” and “the” includes plural references unless the context clearly dictates otherwise. Also, as used in the description herein and throughout the claims that follow, the meaning of “in” includes “in” and “on” unless the context clearly dictates otherwise.
(76) The above description illustrates various examples of the present invention along with examples of how aspects of the present invention may be implemented. The above examples and examples should not be deemed to be the only examples, and are presented to illustrate the flexibility and advantages of the present invention as defined by the following claims. Based on the above disclosure and the following claims, other arrangements, examples, implementations and equivalents may be employed without departing from the scope of the invention as defined by the claims.

Claims

1. A method, comprising: prior to authorizing a client device to access a desktop: registering the client device with an authentication agent of the desktop, the authentication agent obtaining information for an authentication mechanism from the client device, the authentication mechanism configured to authenticate a user on the client device to unlock the desktop, the desktop hosted by a hypervisor running on the client device; receiving an indication on the client device corresponding to the authentication mechanism; determining that the indication is an authorized unlocking indication; upon determining that the indication is an authorized unlocking indication, transmitting client device credentials to a desktop management server, wherein the desktop management server is configured to compare the client device credentials with an enterprise administrative policy to determine whether the client device satisfies the enterprise administrative policy, the enterprise administrative policy comprising the client device being within (i) a given location and (ii) a time of day at which the indication is received; and upon the client device being determined to satisfy the enterprise administrative policy, transmitting the authorized unlocking indication from a desktop agent of the client device to the desktop in a side channel between the desktop agent and the desktop, the side channel authorized for non-display traffic by the desktop management server, wherein the desktop is configured to determine that the client device has been registered with the authentication agent, and determine, based on the authorized unlocking indication, whether the client device is authorized to access the desktop, before unlocking the desktop.
2. The method of claim 1, wherein the client device is provided with a token to access the desktop management server.
3. The method of claim 1, wherein the indication comprises sequentially touching specifically displayed elements on the client device.

4. The method of claim 1, wherein the enterprise administrative policy includes rules associated with one or more of an access level and an access frequency.
5. The method of claim 1, wherein the authentication mechanism includes one or more of a thermogram and gait recognition.
6. The method of claim 1, wherein the indication includes a sequence of a plurality of related events that satisfies the enterprise administrative policy.
7. The method of claim 1, further comprising: receiving the indication on a smart watch; and transmitting the indication from a smart phone to the desktop to unlock the desktop.
8. The method of claim 1, wherein authenticating the user on the client device includes receiving authentication data comprising one or more of voice recognition information and facial recognition information.
9. A system, comprising: one or more memories storing computer-executable instructions; and one or more processors operationally coupled to the one or more memories and configured to execute the computer-executable instructions to: prior to authorizing a client device to access a desktop: register the client device with an authentication agent of the desktop, the authentication agent obtaining information for an authentication mechanism from the client device, the authentication mechanism configured to authenticate a user on the client device unlock the desktop, the desktop hosted by a hypervisor running on the client device; receive an indication, on the client device, corresponding to the authentication mechanism; determine that the indication is an authorized unlocking indication; upon determining that the indication is an authorized unlocking indication, transmit client device credentials of the client device to a desktop management server, wherein the desktop management server is configured to compare the client device credentials with an enterprise administrative policy to determine whether the client device satisfies the enterprise administrative policy, the enterprise administrative policy comprising the client device being within (i) a given location and (ii) a time of day at which the indication is received; and upon the client device being determined to satisfy the enterprise administrative policy, transmit the authorized unlocking indication from a desktop agent of the client device to the desktop in a side channel between the desktop agent and the desktop, the side channel authorized for non-display traffic by the desktop management server, wherein the desktop is configured to determine that the client device has been registered with the authentication agent, and determine, based on the authorized unlocking indication, whether the client device is authorized to access the desktop, before unlocking the desktop.
10. The system of claim 9, wherein the client device is provided with a token to access the desktop management server.
11. The system of claim 9, wherein the indication comprises sequentially touching specifically displayed elements on the client device.
12. The system of claim 9, wherein the indication includes a sequence of a plurality of related events that satisfies the enterprise administrative policy.
13. The system of claim 9, wherein the authentication mechanism includes one or more of: a thermogram and gait recognition.
14. The system of claim 9, wherein authenticating a user on a client device includes receiving authentication data comprising one or more of voice recognition information and facial recognition information.
15. A non-transitory computer-storage memory embodied with instructions executable by one or more processors to enable remote authentication of a desktop by a client device, the instructions comprising: prior to authorizing the client device to access a desktop: registering the client device with an authentication agent of the desktop, the authentication agent obtaining information for an authentication mechanism from the client device, the authentication mechanism configured to authenticate a user on the client device to unlock the desktop, the desktop hosted by a hypervisor running on the client device; receiving an indication on the client device corresponding to the

authentication mechanism; determine whether the indication is an authorized unlocking indication; upon determining that the indication is an authorized unlocking indication, transmitting client device credentials to a desktop management server, wherein the desktop management server is configured to compare the client device credentials with an enterprise administrative policy to determine, whether the client device satisfies the enterprise administrative policy, the enterprise administrative policy comprising the client device being within (i) a given location and (ii) a time of day at which the indication is received; and upon the client device being determined to satisfy the enterprise administrative policy, transmitting the authorized unlocking indication from a desktop agent of the client device to the desktop in a side channel between the desktop agent and the desktop. the side channel authorized for non-display traffic by the desktop management server, wherein the desktop is configured to determine that the client device has been registered with the authentication agent, and determine, based on the authorized unlocking indication, whether the client device is authorized to access the desktop, before unlocking the desktop.

16. The non-transitory computer-storage memory of claim 15, wherein the client device is provided with a token to access the desktop management server.

17. The non-transitory computer-storage memory of claim 15, wherein the instructions further cause disabling of the client device from locking or unlocking the desktop on condition that the client device credentials are determined to be unauthorized.

18. The non-transitory computer-storage memory of claim 15, wherein the enterprise administrative policy includes rules associated with one or more of an access level and an access frequency.

19. The non-transitory computer-storage memory of claim 15, wherein the authentication mechanism includes one or more of: a thermogram, and gait recognition.

20. The non-transitory computer-storage memory of claim 19, wherein authenticating a user on a client device includes receiving authentication data comprising one or more of voice recognition information and facial recognition information.

21. The method of claim 1, wherein the hypervisor is a Type 2 hypervisor.

22. The method of claim 1, further comprising: receiving, at the desktop agent, a results set from the desktop in the side channel, where the results set includes a result of authenticating the user to unlock the desktop.